DILLI RAM ACHARYA

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Career Objective

Aspiring researcher with a background in Agricultural Engineering and interdisciplinary research experience in electrochemical sensing, biomass-derived nanomaterials, and environmental monitoring, seeking opportunities to contribute to innovative research and development in academia or industry. Interested in advancing work on biochar-based biosensors, electrochemical sensing platforms, sustainable energy storage systems such as supercapacitors, and resource recovery technologies including wastewater treatment, critical mineral recovery, and battery recycling. Focused on integrating functional materials, electrochemical techniques, and data-driven approaches to develop sustainable solutions for challenges in environmental monitoring, precision agriculture, and resource-efficient technologies.

Research Interests

- Electrochemical Biosensors and Chemical Sensors
- Wastewater Treatment and Environmental Remediation
- Biochar and Biomass-Derived Functional Materials
- Supercapacitors and Sustainable Energy Storage
- Environmental Monitoring and Heavy Metal Detection
- Sustainable Materials for Battery Recycling and Resource Recovery
- Biomass Valorization and Pyrolysis Technologies
- Machine Learning Applications in Biomass and Sensor Systems
- Sustainable Materials for Agricultural and Environmental Applications

Education

Nanjing Agricultural University, Nanjing, China July 2027
Masters of Engineering, Agricultural Mechanization Engineering *Percentage: 87.9%*

Relevant Coursework: New Energy Utilization and Development, Agricultural Ecology and Environmental Engineering, Agricultural Environment Control Engineering, Special Research Topics on Agricultural Engineering, Higher Agricultural Mechanics, Seminar Discussion, Writing Scientific Papers and Making Presentations in English.

Thesis: Development of Seaweed-Derived Biochar-Modified Sensor for electrochemical detection of heavy metals and nitrogen pollutants (Proposed).

Institute of Engineering, Purwanchal Campus (TU), Dharan, Nepal July 2022
Bachelor of Engineering, Agricultural Engineering *Percentage: 74.5%*

Relevant Coursework: Hydrology and Agricultural Meteorology, Soil and Water Conservation Engineering, Irrigation and Drainage Engineering, Climate Change and Adaptation Measures, Remote Sensing and GIS, Probability and Statistics, Agricultural Mechanization.

Thesis: Economic Analysis of Custom Hiring Center: A Case Study on Dipjyoti Krishi Yantrikaran Custom Hiring Center at Belaka-5, Udayapur District of Nepal.

PUBLICATIONS

Journal paper

- Shen, Y., Li, K., Xue, Y., Ma, C., Cheng, J., Li, M., & **Acharya, D. R.** (2026) Microwave-driven dual-role boron engineering spatially couples in-plane pentagonal defects and edge boron-nitrogen in biocarbon for enrichment-free simultaneous Zn2+/NO2- electrosensing. *Chemical Engineering Journal* (Under Review)
- **Acharya, D.R.**, Li, K., Shen, Y., Wang, D., & Muhammed, I.B (2027). Chronocatalytic mineral depletion-defect nucleation cascade in ball-milled Laminaria japonica biochar enables valley-graphitic nitrogen dyads for sensitive Cd^{2+} electrosensing. *Bioresource Technology* (Under Review).
- Rai, S., Chand, J., **Acharya, D.R.**, Poudyal, K.N. (2026).Assessment of Climate Change Impact on Future Water Availability and Irrigation Demand: A Case Study of the Chanda Mohana Irrigation Project, Nepal. *H2Open Journal*, 9(4), 100039 (2026) (Published).
- Shen, Y., Li, K., Xue, Y., Ma, C., Cheng, J., Li, M., & **Acharya, D. R.** (2026). Microwave-coupled molten potassium salt engineering structurally separates boron/nitrogen-biocarbon for signal-amplified multiplexed electrodetection of Cd^{2+}, Pb^{2+} and Hg^{2+} micropollutants. *Journal of Colloid and Interface Science*, 706, 139615 (Published).
- Shen, Y., Li, K., Xue, Y., Ma, C., Cheng, J., Miao, S., & **Acharya, D. R.** (2026). Dual-role boron and oxidized phosphorus in collectively constructing the signal-amplified bio-carbon electroanalysis platforms for simultaneous recognition of sulfite and nitrite in aquaculture. *Sensors and Actuators B: Chemical*, 447, 138906 (Published).

TECHNICAL SKILLS

Electrochemical Techniques

* Cyclic Voltammetry (CV)
* Differential Pulse Voltammetry (DPV)
* Square Wave Voltammetry (SWV)
* Electrochemical Impedance Spectroscopy (EIS)
* Electrochemical Sensor Fabrication and Optimization

Materials Synthesis & Processing

* Biochar and Nanobiochar Synthesis
* Biomass Pyrolysis
* Ball Milling Techniques
* Electrode Surface Modification

Material Characterization

* SEM
* FTIR
* XRD
* Raman Spectroscopy
* BET Surface Area Analysis

Computational Chemistry & DFT Analysis

* Gaussian, GaussView, Multiwfn, and VMD
* Density Functional Theory (DFT) Calculations
* Frontier Molecular Orbital Analysis (HOMO-LUMO)
* Electrostatic Potential (ESP) and ALIE Analysis
* Interaction Region Indicator (IRI) Analysis
* Adsorption Energy and Charge Distribution Analysis

Programming & Data Analysis

* Python (basic data analysis and machine learning)

* OriginPro
* Design-Expert
* Microsoft Excel

Machine Learning & Modeling

* Random Forest
* XGBoost
* Artificial Neural Networks (ANN)
* Response Surface Methodology (RSM)

Research & Professional Skills

* Scientific Writing and Literature Review
* Experimental Design and Optimization
* Academic Presentation and Communication

EXPERIENCE

Nepal Agricultural Engineering Students Society (NAESS) | *Vice Secretary* 2019 – 2020
Assists secretary & team in administrative tasks and organization

DELTA 2.0 | *Financial Coordinator* 2020 – 2021
Manages budgets, expenses, and financial reporting

Clamphook Academy | *Instructor* 2022 – 2024
Teaches mathematical concepts and problem-solving skills to class 12 students

Apex Academy | *Instructor* 2022 – 2024
Routinely tutoring 12 passed students in mathematics

Nepal Kasthamadap College | *Tutor* 2022 – 2023
Routinely tutoring class 11, 12 & BBA students in mathematics

National College of Science(NIST) | *Tutor* 2022 – 2023
Routinely tutoring 11 & 12 students in mathematics

SELECTED COURSES

Master's Courses

- Heat and Mass Transfer
- Agricultural Sensing and Information Acquisition Technology
- Agricultural Environment Control Engineering
- Matrix Theory
- New Energy Utilization and Development
- Chinese Language and Culture
- Seminar Discussion
- Higher Agricultural Mechanics
- Special Research Topic on Agricultural Engineering
- Agricultural Ecology and Environment Engineering
- Engineering Testing Technology
- Drying Principle
- Studies on History of Science and Technology
- Writing Scientific Paper and Making Presentation in English
- Higher Engineering Thermodynamics
- Higher Engineering Mechanics (Mechanical Vibration)

GRANTS & AWARDS

China Scholarship Council (CSC) Nanjing, China
Fully Scholarship for my Master's degree 2024 September to 2027 July

Winner, University-Level Essay Writing Competition *May 2025*
Event: Perceiving China 2025 Program
Organizer: College of International Education (COIE), Nanjing Agricultural University

LEADERSHIP & EXTRACURRICULAR ACTIVITES

Perceiving China 2025 Program (Yangzhou Visit) *May 2025*
Organizer: College of International Education (COIE), Nanjing Agricultural University

Qixia Mountain Cultural and Natural Exploration Activity *September 2025*
Organizer: School of Life Sciences & School of International Education, Nanjing Agricultural University

University Sports Event (Marching Program, Group Running, Shot Put) *October 2025*
Organizer: College of International Education (COIE), Nanjing Agricultural University

International Cultural Program (Representing Nepal) *November 2025*
Organizer: College of International Education (COIE), Nanjing Agricultural University

Graduate Hiking Competition (Purple Mountain) *April 2026*
Organizer: Nanjing Agricultural University

2026 The Road and Belt Youth Sports Exchange Week (Jiangsu) *May 2026*
Organizer: Jiangsu Provincial Sports and Educational Organizations

PROFESSIONAL MEMBERSHIP

- Nepal Engineers Association (NEA)
- Nepalese Society of Agricultural Engineers (NSAE)

LANGUAGES

- Nepali (Native)
- English (Fluent, Academic/Technical proficiency)
- Hindi (Fluent)
- Chinese (Basic)

HOBBIES

- Badminton and Recreational Sports
- Weight Lifting, Gym
- Hiking and Nature Exploration
- Cultural Exchange and International Networking
- Content Creation

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